

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1534

Yorkshire 138 kV
October 21, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1534. The Interconnection Customer proposes a 110.8 MW solar photovoltaic facility interconnecting at the Yorkshire 138 kV substation in the Pennsylvania area. The request was received on July 28, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 531 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,784,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: PAXON – S LAKE GENEVA 138 kV
Contingency: TALENENERGY MARTINS CREEK LLC HARRISBURG 500/345kV Transformer
Loading: 176.4% (Rating: 206 MVA)
DFAX: 22.8%
Pre-project loading: 164.1%
2. Facility: S LAKE GENEVA – HUMMELSTOWN 138 kV
Contingency: TALENENERGY MARTINS CREEK LLC HARRISBURG 500 kV/345 kV XFMR
Loading: 144.3% (Rating: 230 MVA)
DFAX: 22.8%
Pre-project loading: 133.4%
3. Facility: RIV BEND SUBSTATION – HUMMELSTOWN 138 kV
Contingency: TALENENERGY MARTINS CREEK LLC HARRISBURG 500 kV/345 kV Transformer Bank
Loading: 133.6% (Rating: 239 MVA)
DFAX: 22.8%
Pre-project loading: 123.1%
4. Facility: RIV BEND – GARBERVILLE 138 kV
Contingency: TALENENERGY MARTINS CRK LLC HARRISBURG SUBSTATION 500-345 kV Transformer
Loading: 123.8% (Rating: 248 MVA)
DFAX: 22.8%
Pre-project loading: 113.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
S LAKE GENEVA – PAXON 138 kV	Rebuild line with higher-capacity conductor	\$1,543,000	77% (\$1,185,000)
SOUTH LAKE GENEVA – HUMMELSTOWN 138 kV line	Rebuild line with higher-capacity conductor	\$5,288,000	77% (\$4,062,000)
RIVR BEND – HUMMELSTOWN 138 kV	Rebuild line with higher-capacity conductor	\$1,927,000	77% (\$1,480,000)
GARBERVILLE – RIV BEND 138 kV	Rebuild line with higher-capacity conductor	\$1,376,000	77% (\$1,057,000)
	Total allocated to project		\$7,784,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.